

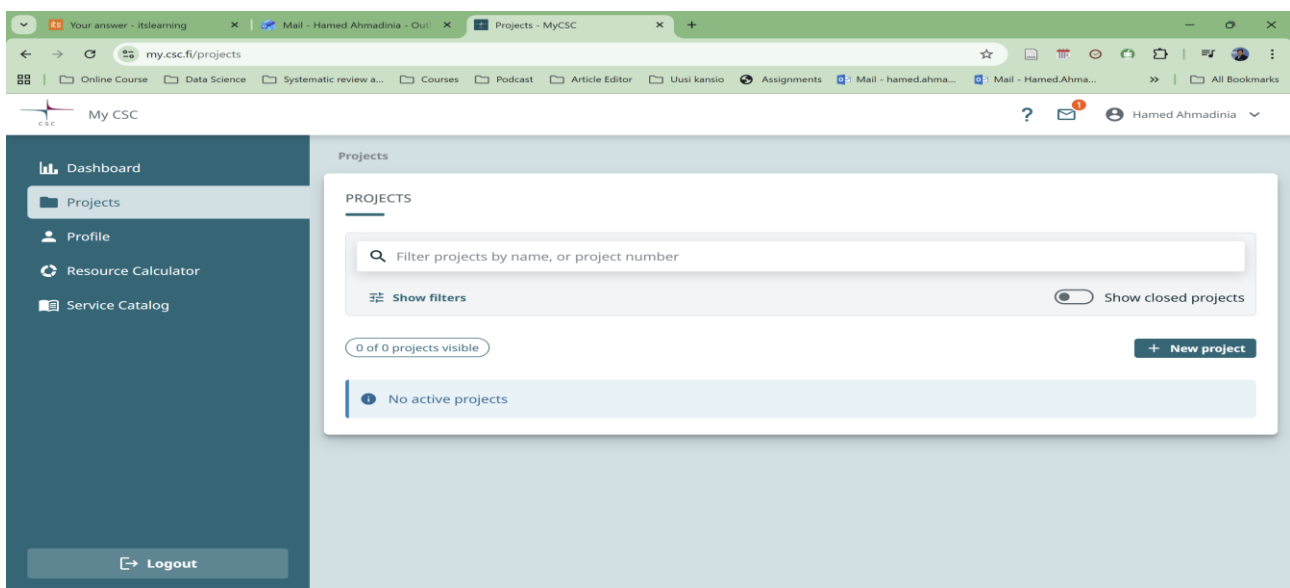
Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Introduction

The purpose of this assignment was to gain hands-on experience with Infrastructure as a Service (IaaS) by creating and managing a virtual server on CSC Pouta. The task included setting up a project, launching an Ubuntu 24.04 virtual machine, configuring secure access, installing essential software, and making a web server accessible through a public IP address. This report describes all the steps I followed, from the initial setup to the final verification.

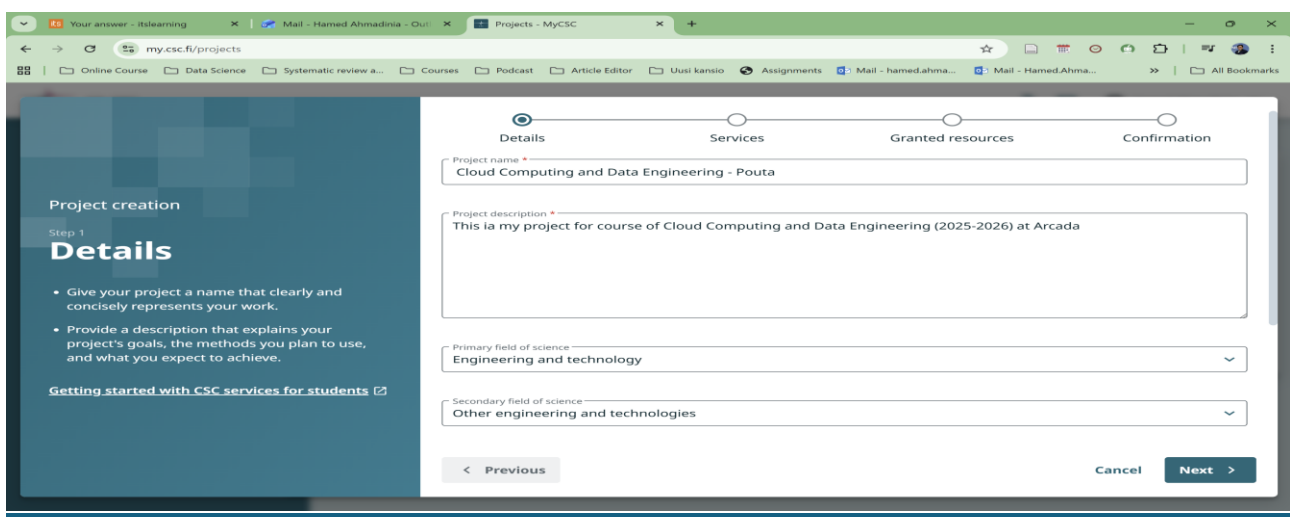
Step 1 – Logging in to MyCSC

I logged in to the MyCSC portal using my CSC credentials to access cloud services and manage projects.



Step 2 – Creating a New Project

I created a new project in MyCSC and gave it a clear name related to the course *Cloud Computing and Data Engineering*. I added a short description explaining that the project is used for coursework.



Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 3 – Selecting Field of Science

I selected **Engineering and Technology** as the primary field of science and an appropriate secondary field, as required during project creation.

The screenshot shows the 'my.csc.fi/projects' web interface. On the left, a sidebar titled 'Project creation' indicates 'Step 2: Services'. The main content area shows a progress bar with four steps: Details (checked), Services (active), Granted resources, and Confirmation. Below the progress bar, the heading 'Select services for the project' is followed by a list of services with checkboxes:

- ☐ Allas: Object storage environment for storing and sharing data
- ☒ cPouta: Virtual machines for web services & software development
- ☐ Puhti: Core based supercomputing environment
- ☐ Pukki DBaaS: A self service on-demand database service
- ☐ Skip service and resource selection: It is possible to enable services later.

At the bottom, there are buttons for '< Previous', 'Cancel', and 'Next >'.

Step 4 – Selecting Services

I selected **cPouta** as the service for the project, since it provides virtual machines and supports Infrastructure as a Service (IaaS).

The screenshot shows the 'my.csc.fi/projects' web interface at 'Step 3: Granted resources'. The progress bar now shows 'Details' and 'Services' as completed steps, with 'Granted resources' as the active step. The main content area displays a 'Cloud Resource Package' section, stating: 'The project will be granted 30 000 cloud billing units' and 'Selected services consuming these resources: cPouta'. At the bottom, there are buttons for '< Previous', 'Cancel', and 'Next >'.

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 5 – Reviewing Granted Resources

After selecting the service, the system showed the granted cloud resources (billing units). I reviewed the information and confirmed the setup.

The screenshot shows a web browser window with the URL `my.csc.fi/projects`. The page is titled "Project creation" and "Step 4 Confirmation". It contains a list of terms and conditions, including a warning about export restrictions and a list of links for more information. A checkbox is checked, indicating acceptance of the terms. The page has "Previous", "Cancel", and "Submit" buttons.

Project creation
Step 4
Confirmation

- Please accept the terms and conditions and submit the form.

More information about the current export restrictions:

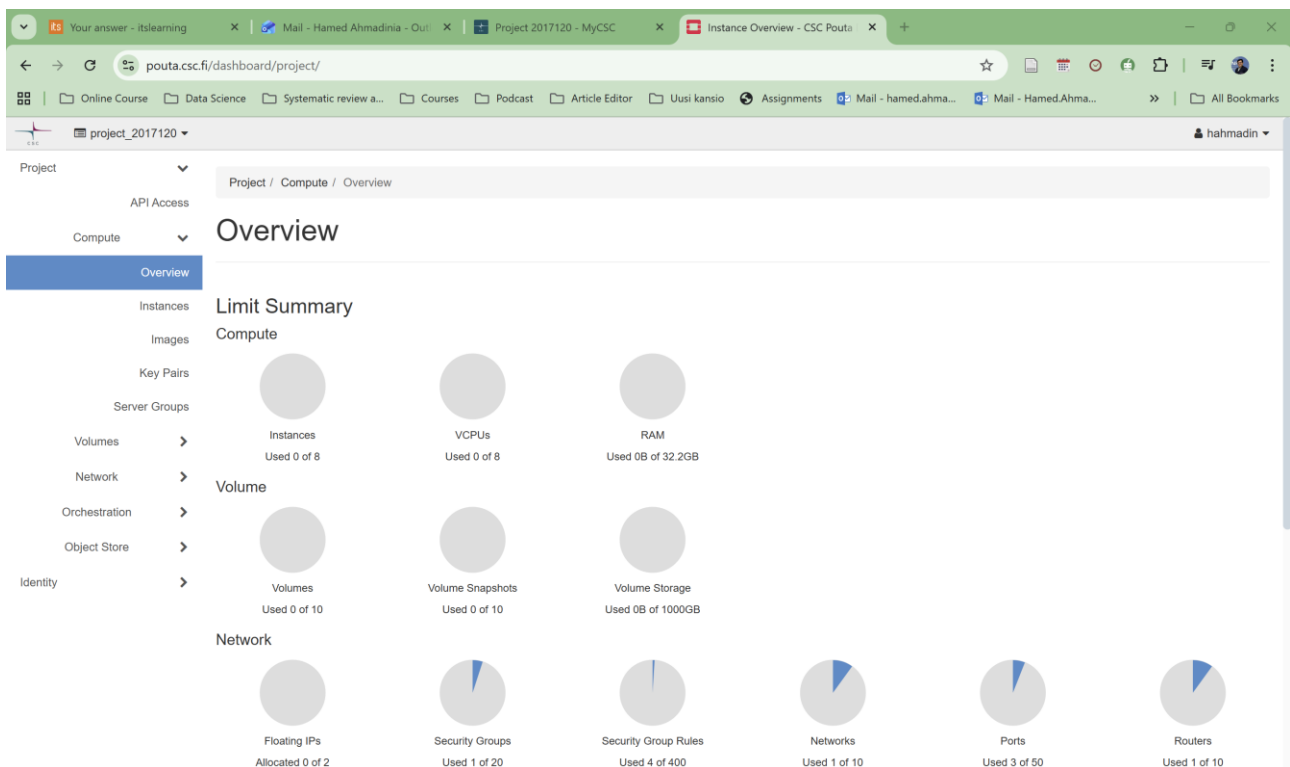
- You do not circumvent the above-mentioned prohibitions by technical means.
- If you are uncertain whether export restrictions or sanctions apply to your research project, we recommend to contact the Finnish home organization of your project or CSC Service Desk.
- Finnish law on dual-use items export control 500/2024 (Laki kaksikäyttötuotteiden vientivalvonnasta, j...**
- Regulation (EU) 2021/821 of the European Parliament and of the Council** for the control of exports, brokering, technical assistance, transit and transfer of dual-use items (recast), especially the Annex 1
- Sanctions by countries (Ministry for Foreign Affairs, in Finnish)**, **Export Control (Ministry for Foreign Affairs, in English)**. In particular, sanctions against Russia (Ukraine) are the most important consolidated versions of Council Regulation (EU) 833/2014 and (EU) 269/2014.
- EU sanctions map on dual-use items**
- Information on US export regulations and sanctions**
- US Resources on Export Controls Implemented in Response to Russia's Invasion of Ukraine**
- US Commerce Control List**
- US Commerce Country Chart**

☒ I have read and accepted [the Export Restrictions and Sanctions](#).

[< Previous](#) [Cancel](#) [Submit](#)

Step 6 – Accepting Terms and Confirming Project

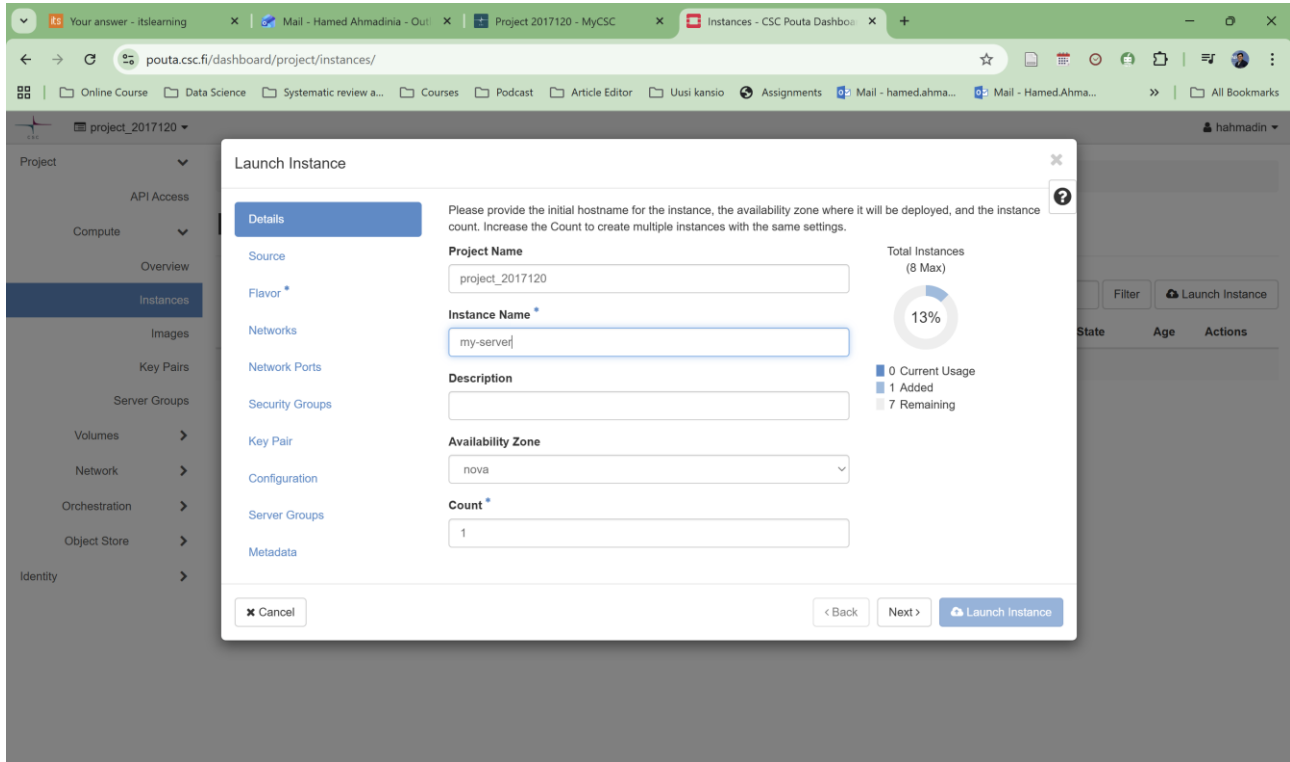
I accepted the export restriction terms and submitted the project creation form. The project was then successfully created.



Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

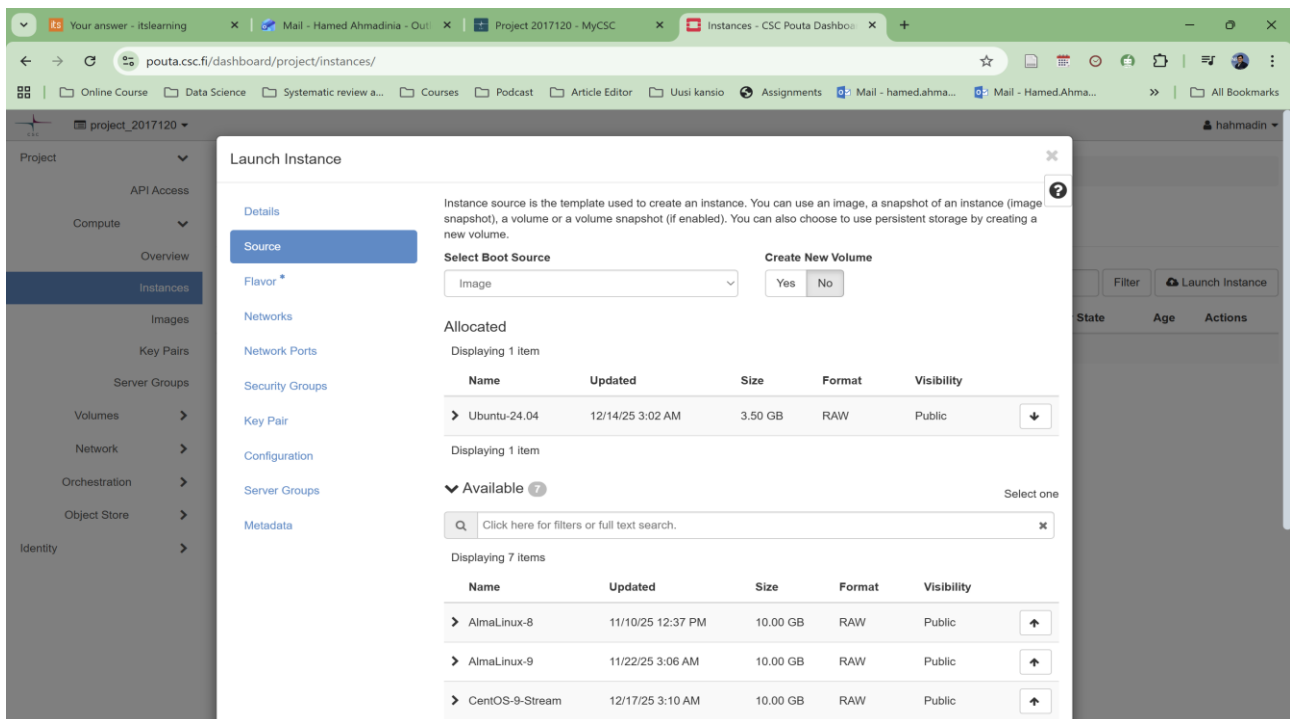
Step 7 – Opening CSC Pouta (OpenStack) Dashboard

After the project was active, I opened the CSC Pouta (OpenStack) dashboard to manage virtual machines and networking.



Step 8 – Launching a New Instance

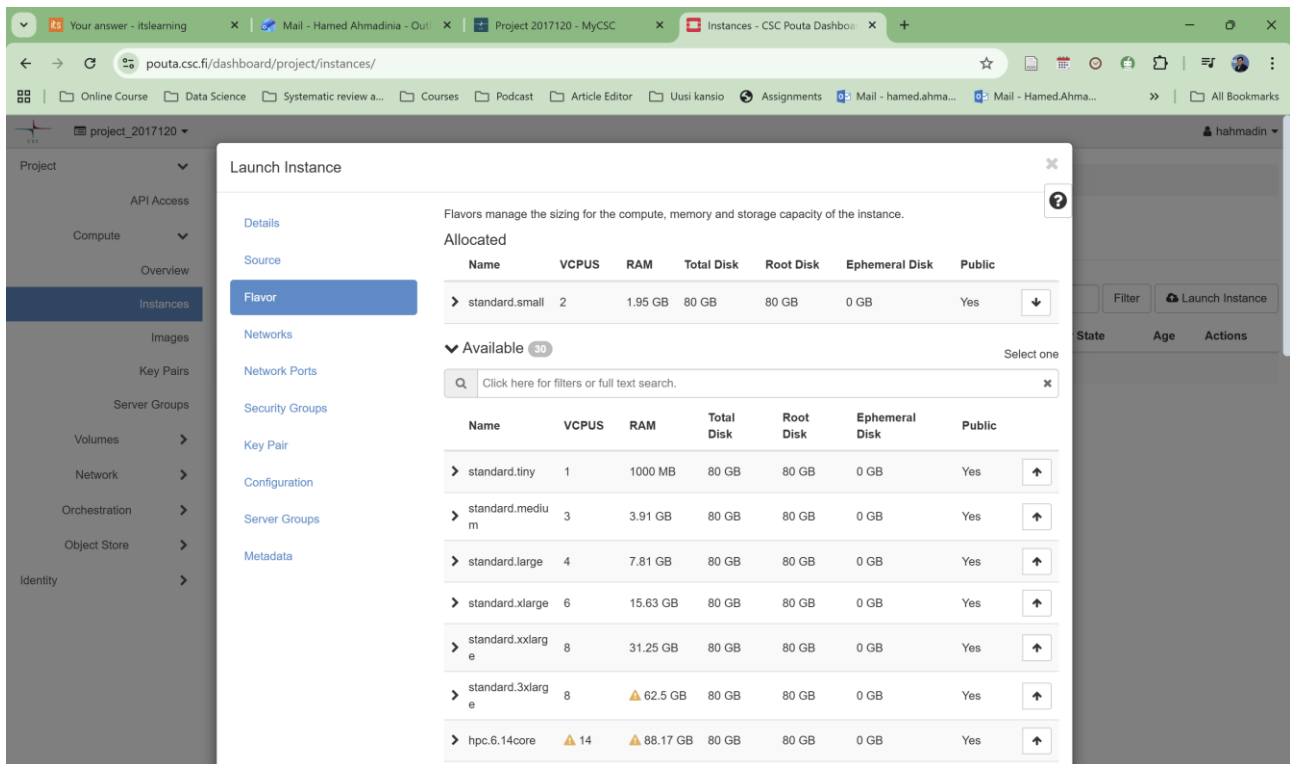
In the OpenStack dashboard, I selected **Launch Instance** to create a new virtual machine.



Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

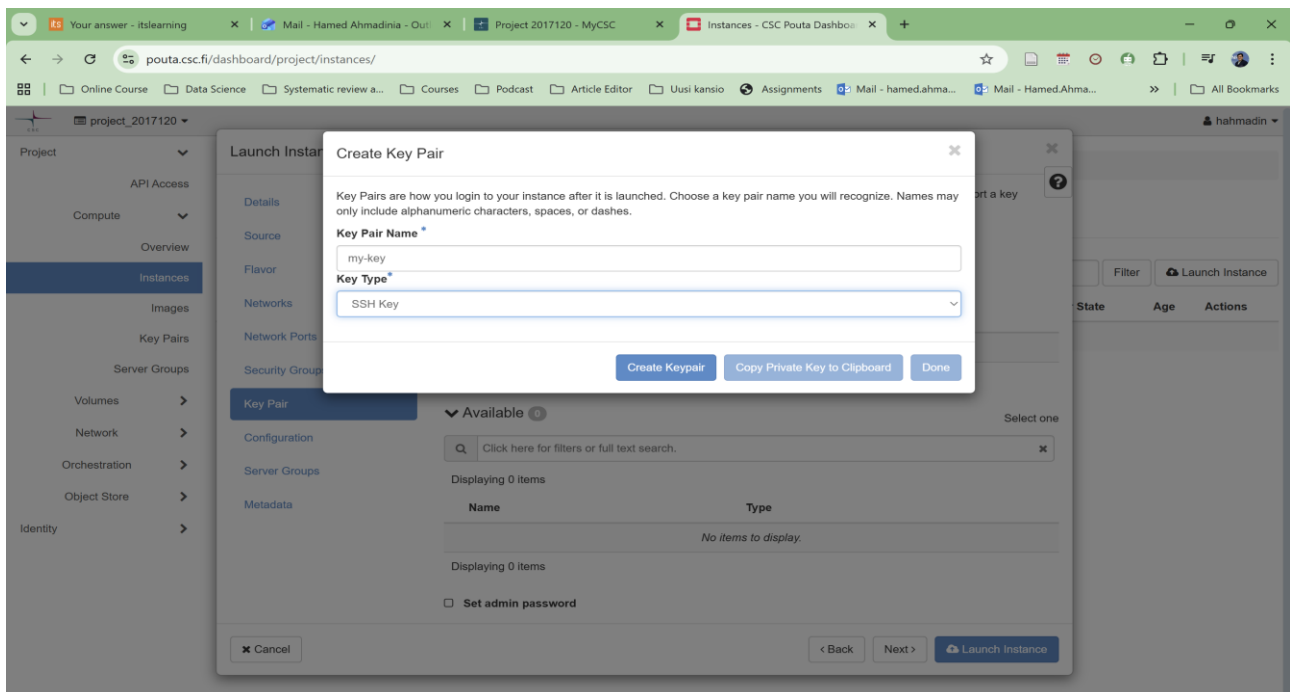
Step 9 – Instance Details

I named the instance my-server, selected the availability zone nova, and kept the instance count as one.



Step 10 – Selecting the Image

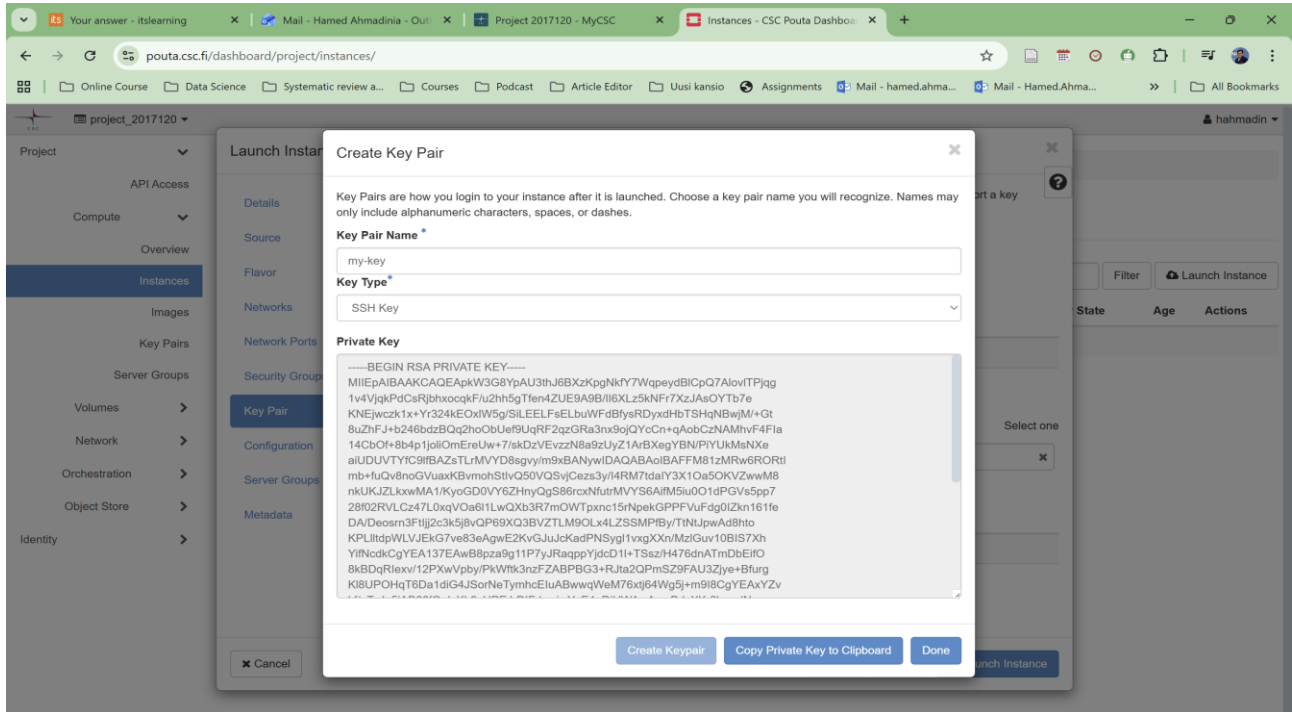
I selected **Ubuntu 24.04** as the operating system image for the virtual machine.



Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

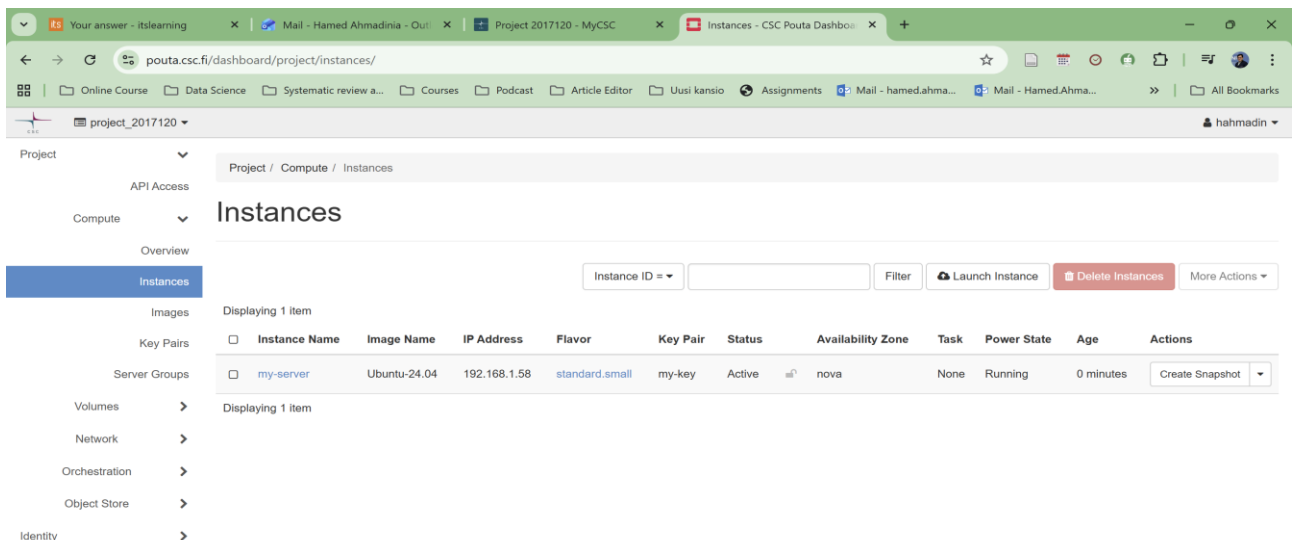
Step 11 – Selecting the Flavor

I chose the standard.small flavor, which provides sufficient CPU, memory, and disk space for the assignment tasks.



Step 12 – Creating an SSH Key Pair

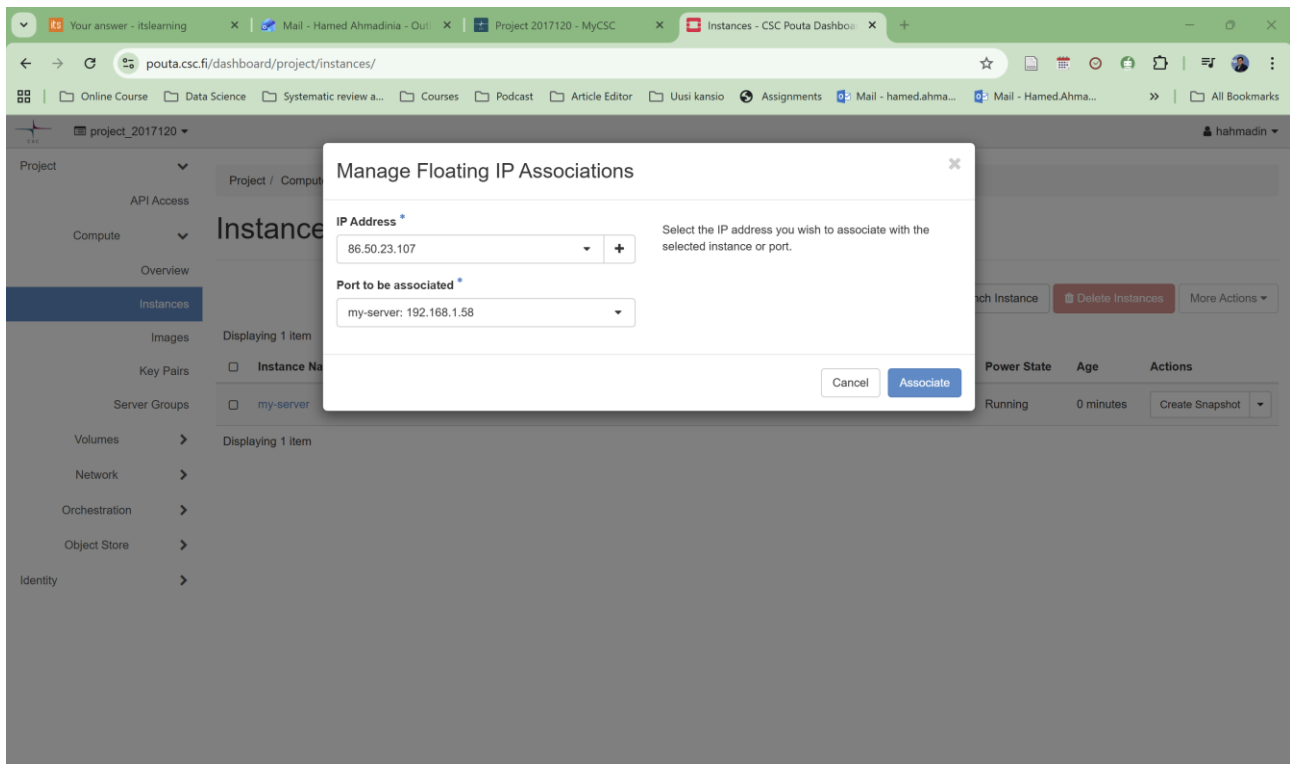
I created a new SSH key pair during the instance setup and downloaded the private key file to my local computer. This key is required for secure login to the server.



Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 13 – Launching the Instance

After reviewing all settings, I launched the instance. The virtual machine started successfully and appeared as running in the dashboard.



Step 14 – Assigning a Floating IP

I allocated a floating IP address and associated it with the instance. This allowed the server to be accessed from outside the cloud network.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\> ssh -i my-key.rsa ubuntu@192.168.1.58
ssh: connect to host 192.168.1.58 port 22: Connection timed out
PS D:\> ssh -i my-key.rsa ubuntu@86.50.23.107
ssh: connect to host 86.50.23.107 port 22: Connection timed out
PS D:\> ssh -i my-key.rsa ubuntu 86.50.23.17
ssh: Could not resolve hostname ubuntu: No such host is known.
PS D:\> ssh -i my-key.rsa ubuntu@86.50.23.17
ssh: connect to host 86.50.23.17 port 22: Connection refused
PS D:\> ls

Directory: D:\

Mode                LastWriteTime         Length Name
----                -
-a-----         12/20/2025   3:01 PM           1706 my-key.rsa

PS D:\>
```

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 15 – Initial SSH Connection Attempts

I attempted to connect to the server using SSH. Initial connection attempts failed because SSH access was not yet allowed in the security group.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS D:\> ssh -i my-key.rsa ubuntu@192.168.1.58
ssh: connect to host 192.168.1.58 port 22: Connection timed out
PS D:\> ssh -i my-key.rsa ubuntu@86.50.23.107
ssh: connect to host 86.50.23.107 port 22: Connection timed out
PS D:\> ssh -i my-key.rsa ubuntu 86.50.23.17
ssh: Could not resolve hostname ubuntu: No such host is known.
PS D:\> ssh -i my-key.rsa ubuntu@86.50.23.17
ssh: connect to host 86.50.23.17 port 22: Connection refused
PS D:\> ls

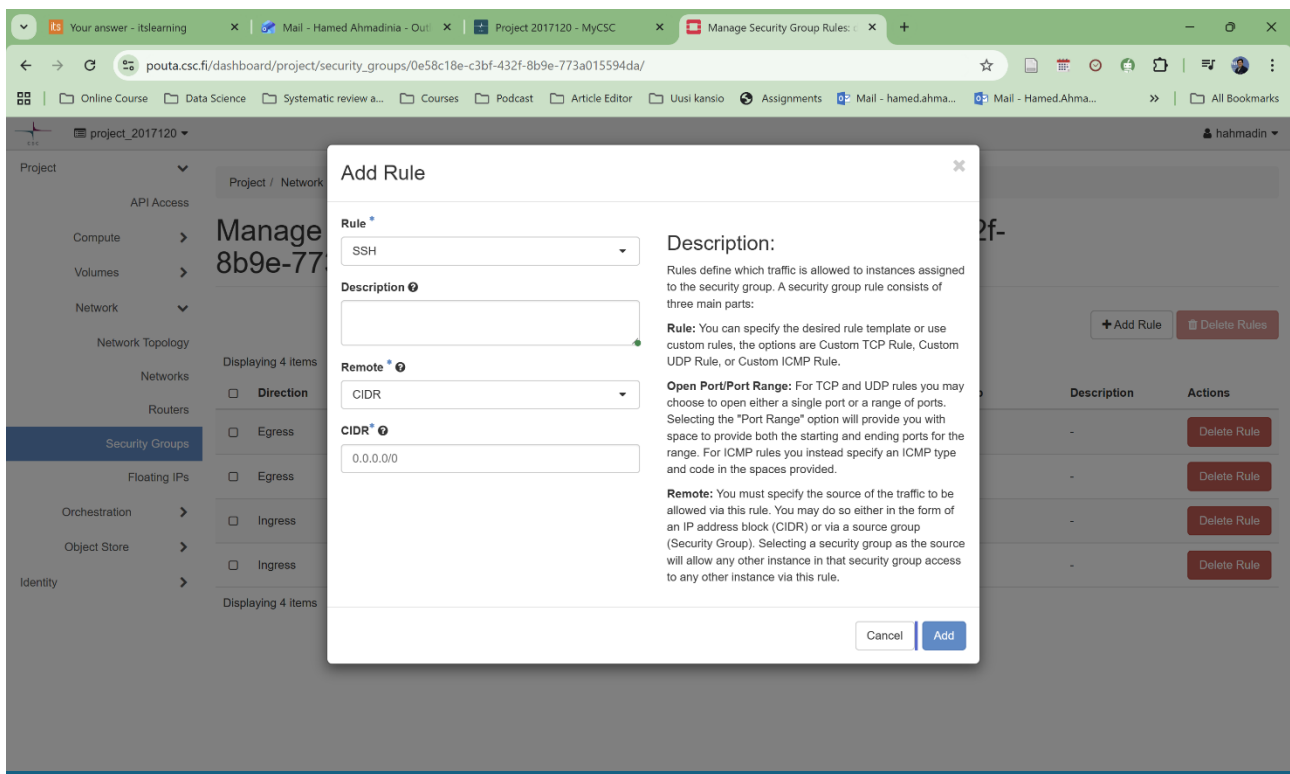
Directory: D:\

Mode                LastWriteTime         Length Name
----                -
-a-----         12/20/2025   3:01 PM           1706 my-key.rsa

PS D:\> ssh -i .\my-key.rsa ubuntu@86.50.23.107
ssh: connect to host 86.50.23.107 port 22: Connection timed out
PS D:\> |
```

Step 16 – Configuring Security Group for SSH

I edited the default security group and added an ingress rule allowing **SSH (port 22)** from 0.0.0.0/0. After this, SSH connections worked correctly.



Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 17 – Successful SSH Login

Using Windows PowerShell and the private key, I logged in to the server as the ubuntu user via SSH.

```
ubuntu@my-server: ~  
System information as of Sat Dec 20 14:21:02 UTC 2025  
  
System load: 0.0      Processes:            108  
Usage of /:  2.1% of 76.45GB   Users logged in:     0  
Memory usage: 9%      IPv4 address for ens3: 192.168.1.58  
Swap usage:  0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@my-server:~$ |
```

Step 18 – Verifying the Operating System

I verified the operating system and kernel version using the `uname -a` command, confirming that Ubuntu 24.04 was running.

```
ubuntu@my-server: ~  
Usage of /:  2.1% of 76.45GB   Users logged in:     0  
Memory usage: 9%      IPv4 address for ens3: 192.168.1.58  
Swap usage:  0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@my-server:~$ uname -a  
Linux my-server 6.8.0-90-generic #91-Ubuntu SMP PREEMPT_DYNAMIC Tue Nov 18 14:14:30 UTC 2025 x86_64 x86_64 x86_64  
GNU/Linux  
ubuntu@my-server:~$ |
```

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 19 – Checking Administrative Access

I tested administrative privileges by using sudo to switch to the root user and confirmed that I had full administrative access.

```
root@my-server: /home/ubur  X + v
Memory usage: 9%          IPv4 address for ens3: 192.168.1.58
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@my-server:~$ uname -a
Linux my-server 6.8.0-90-generic #91-Ubuntu SMP PREEMPT_DYNAMIC Tue Nov 18 14:14:30 UTC 2025 x86_64 x86_64 x86_64
GNU/Linux
ubuntu@my-server:~$ sudo -s
root@my-server:/home/ubuntu#
```

Step 20 – Exiting Root and Returning to User

I exited the root shell and returned to the normal ubuntu user, confirming correct privilege handling.

```
root@my-server: /home/ubur  X + v
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@my-server:~$ uname -a
Linux my-server 6.8.0-90-generic #91-Ubuntu SMP PREEMPT_DYNAMIC Tue Nov 18 14:14:30 UTC 2025 x86_64 x86_64 x86_64
GNU/Linux
ubuntu@my-server:~$ sudo -s
root@my-server:/home/ubuntu# whoami
root
root@my-server:/home/ubuntu#
```

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 21 – Confirming System State

I reviewed the system status message shown after login, confirming disk usage, memory usage, and network configuration.

```
ubuntu@my-server: ~  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@my-server:~$ uname -a  
Linux my-server 6.8.0-90-generic #91-Ubuntu SMP PREEMPT_DYNAMIC Tue Nov 18 14:14:30 UTC 2025 x86_64 x86_64 x86_64  
GNU/Linux  
ubuntu@my-server:~$ sudo -s  
root@my-server:/home/ubuntu# whoami  
root  
root@my-server:/home/ubuntu# exit  
exit  
ubuntu@my-server:~$
```

Step 22 – Updating the System

I updated the package list and installed all available updates using:

- sudo apt update
- sudo apt upgrade -y

This ensured the system was fully up to date.

```
ubuntu@my-server: ~  
Get:27 http://nova.clouds.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [488 B]  
Get:28 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [64.0 kB]  
Get:29 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]  
Get:30 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7340 B]  
Get:31 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]  
Get:32 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [33.0 kB]  
Get:33 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.9 kB]  
Get:34 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]  
Get:35 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1444 B]  
Get:36 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]  
Get:37 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]  
Get:38 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]  
Get:39 http://nova.clouds.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]  
Get:40 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:41 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1391 kB]  
Get:42 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [225 kB]  
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]  
Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9504 B]  
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [916 kB]  
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [207 kB]  
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.5 kB]  
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.4 kB]  
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2286 kB]  
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [523 kB]  
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]  
Get:52 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]  
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5956 B]  
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]  
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]  
Fetched 39.3 MB in 31s (1252 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
26 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu@my-server:~$
```

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 23 – Installing Essential Software

I installed additional required tools:

- **htop** for system monitoring
- **lynx** for browsing the web from the command line

Both programs were installed and tested successfully.

```
ubuntu@my-server: ~  
Setting up systemd-resolved (255.4-1ubuntu8.12) ...  
Setting up systemd-sysv (255.4-1ubuntu8.12) ...  
Setting up libnss-systemd:amd64 (255.4-1ubuntu8.12) ...  
Setting up libpam-systemd:amd64 (255.4-1ubuntu8.12) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...  
Processing triggers for initramfs-tools (0.142ubuntu25.5) ...  
update-initramfs: Generating /boot/initrd.img-6.8.0-90-generic  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
Restarting services...  
systemctl restart multipathd.service packagekit.service polkit.service rsyslog.service ssh.service udisks2.servic  
e  
  
Service restarts being deferred:  
systemctl restart ModemManager.service  
/etc/needrestart/restart.d/dbus.service  
systemctl restart getty@tty1.service  
systemctl restart serial-getty@ttyS0.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
ubuntu @ session #11: apt[1805], sshd[1221]  
ubuntu @ user manager service: systemd[1226]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@my-server:~$
```

Step 24 – Setting a Password for Console Login

I set a password for the ubuntu user using `sudo passwd ubuntu`. This allowed me to log in via the OpenStack web console in addition to SSH.

```
ubuntu@my-server: ~  
Preparing to unpack .../mailcap_3.70+nmulubuntu1_all.deb ...  
Unpacking mailcap (3.70+nmulubuntu1) ...  
Selecting previously unselected package lynx.  
Preparing to unpack .../lynx_2.9.0rel.0-2build2_amd64.deb ...  
Unpacking lynx (2.9.0rel.0-2build2) ...  
Setting up bzip2 (1.0.8-5.1build0.1) ...  
Setting up htop (1.3.0-1) ...  
Setting up lynx-common (2.9.0rel.0-2build2) ...  
Setting up lynx (2.9.0rel.0-2build2) ...  
update-alternatives: using /usr/bin/lynx to provide /usr/bin/www-browser (www-browser) in auto mode  
Setting up mailcap (3.70+nmulubuntu1) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
Restarting services...  
  
Service restarts being deferred:  
/etc/needrestart/restart.d/dbus.service  
systemctl restart getty@tty1.service  
systemctl restart serial-getty@ttyS0.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
ubuntu @ session #11: sshd[1221]  
ubuntu @ user manager service: systemd[1226]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@my-server:~$
```

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 25 – Installing the Apache Web Server

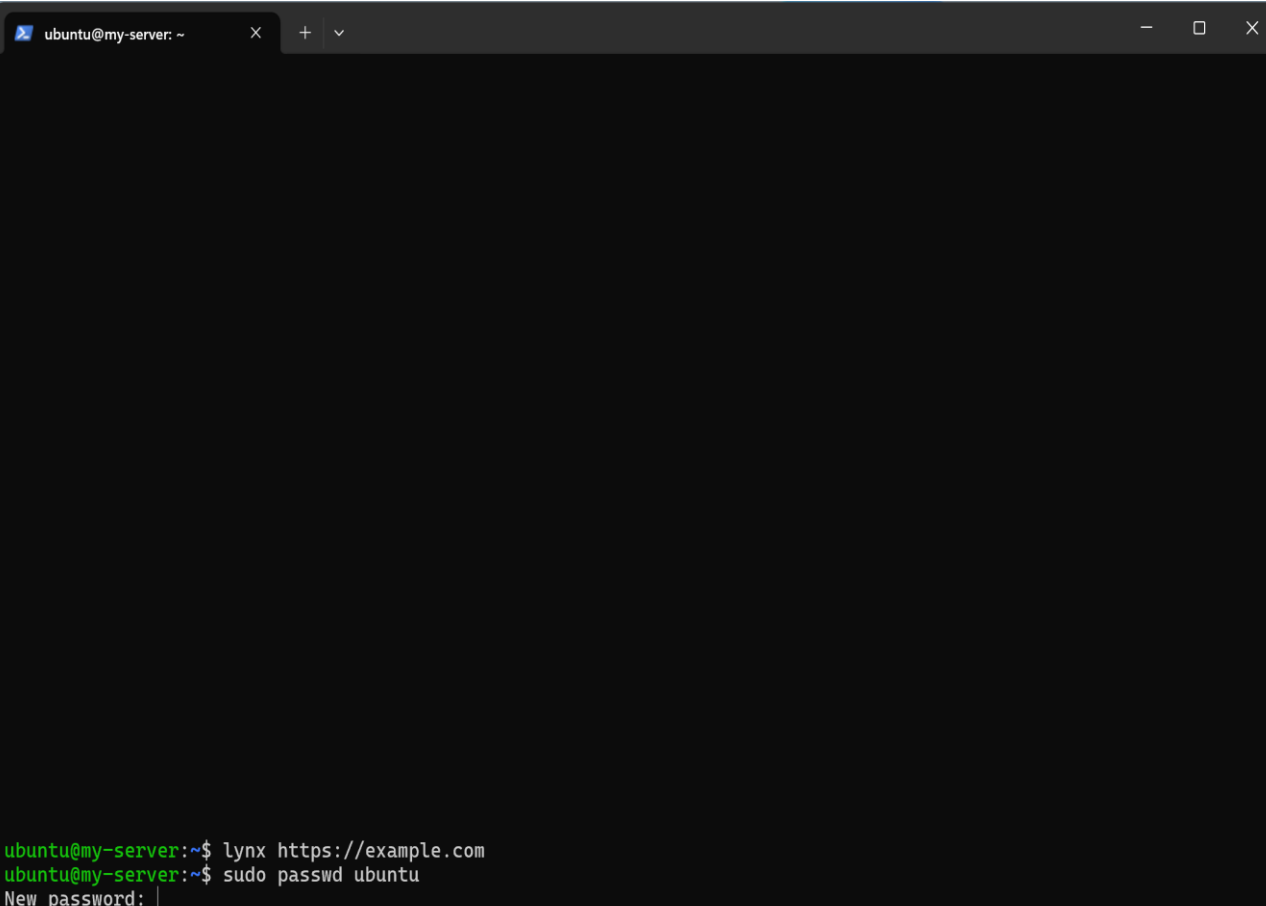
I installed the Apache web server using: `sudo apt install apache2`

After installation, I verified that the Apache service was running.

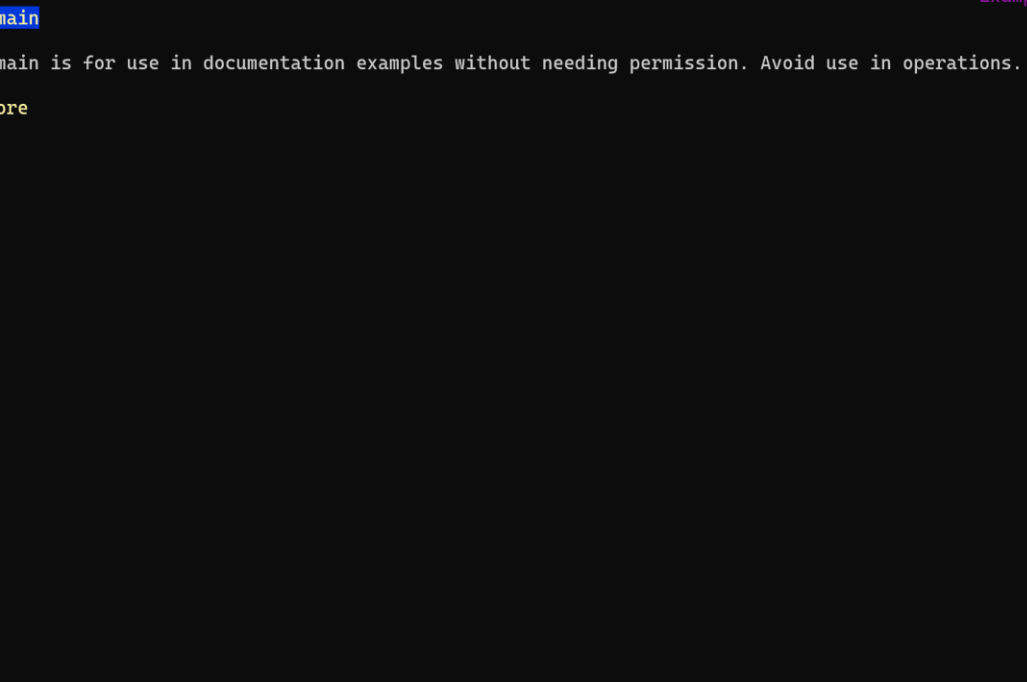
```
ubuntu@my-server: ~  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
*cpu menu preset * 14:45:21 2000ms +  
Xeon (C 2.0 GHz  
CPU 0%  
C0 1%  
C1 0%  
LAV: 0.21 0.15 0.06  
up 01:42:11  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
*mem disks io *proc filter reverse tree < cpu lazy >  
Total: 1.87 GiB root 76.4 GiB  
Used: 325 MiB IO% 3% 1.91 GiB  
Available: 1.55 GiB boot 880 MiB  
IO% 14% 125 MiB  
Cached: 733 MiB efi 104 MiB  
IO% 6% 6.10 MiB  
Free: 930 MiB  
IO% 49%  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
*net sync auto zero <b ens3 n>  
10K  
download  
▼ 55.7 MiB/s (445 Mibps)  
▼ Top: (445 Mibps)  
▼ Total: 3.43 GiB  
▲ 929 KiB/s (7.25 Mibps)  
▲ Top: (7.25 Mibps)  
▲ Total: 90.2 MiB  
upload  
10K  
↑ select ↓ info ↵ terminate ↵ kill ↵ signals 0/118
```

```
ubuntu@my-server: ~  
Enabling conf localized-error-pages.  
Enabling conf other-vhosts-access-log.  
Enabling conf security.  
Enabling conf serve-cgi-bin.  
Enabling site 000-default.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.  
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.  
Processing triggers for ufw (0.36.2-6) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
Restarting services...  
  
Service restarts being deferred:  
/etc/needrestart/restart.d/dbus.service  
systemctl restart getty@tty1.service  
systemctl restart serial-getty@ttyS0.service  
systemctl restart systemd-logind.service  
systemctl restart unattended-upgrades.service  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
ubuntu @ session #11: sshd[1221]  
ubuntu @ user manager service: systemd[1226]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@my-server:~$
```

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

A terminal window titled 'ubuntu@my-server: ~' with standard window controls. The terminal shows the execution of 'lynx https://example.com' and 'sudo passwd ubuntu'. The 'passwd' command prompts for a new password, which is entered as '12345678'.

```
ubuntu@my-server:~$ lynx https://example.com
ubuntu@my-server:~$ sudo passwd ubuntu
New password: 12345678
```



```
ubuntu@my-server: ~  
Example Domain  
This domain is for use in documentation examples without needing permission. Avoid use in operations.  
| Learn more  
  
Commands: Use arrow keys to move, '?' for help, 'q' to quit, '<-' to go back.  
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.  
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```


Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Step 26 – Configuring Security Group for Web Traffic

I added ingress rules to the default security group to allow:

- HTTP (port 80)
- HTTPS (port 443)

This enabled web access to the server.

Project / Network / Security Groups / Manage Security Group Rules: default (0e58c18e-c3bf-432f-8b9e-773a015594da)

Success: Successfully added rule: ALLOW IPv4 443/tcp from 0.0.0.0/0

Direction	Ether Type	IP Protocol	Port Range	Remote IP Prefix	Remote Security Group	Description	Actions
Egress	IPv4	Any	Any	0.0.0.0/0	-	-	Delete Rule
Egress	IPv6	Any	Any	:::/0	-	-	Delete Rule
Ingress	IPv4	Any	Any	-	default	-	Delete Rule
Ingress	IPv4	TCP	22 (SSH)	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv4	TCP	80 (HTTP)	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv4	TCP	443 (HTTPS)	0.0.0.0/0	-	-	Delete Rule
Ingress	IPv6	Any	Any	-	default	-	Delete Rule

Step 27 – Testing the Web Server

Finally, I opened the floating IP address in a web browser. The **Apache2 Ubuntu Default Page** loaded successfully, confirming that the web server was accessible from the internet.

Apache2 Default Page

Ubuntu **It works!**

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/  
|-- apache2.conf  
|-- ports.conf  
|-- mods-enabled/  
|   |-- *.load  
|   |-- *.conf  
|-- conf-enabled/  
|   |-- *.conf  
|-- sites-enabled/  
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/` subdirectories. *Thou should be aware that some modules are placed in a `disabled` state by default.*

Assignment 1: IaaS – My Own Server (CSC Pouta) Hamed Ahmadinia

Conclusion

Through these 27 steps, I successfully set up and managed an Ubuntu 24.04 virtual server using CSC Pouta. I configured secure access with SSH keys, assigned a floating IP, managed security group rules, updated the system, installed essential software, and deployed a working web server. The final browser test confirmed that the server was correctly configured and publicly accessible, fulfilling all the requirements of Assignment 1.

